User manual / Environment setup

Setting Up the Development Environment

This document provides step-by-step instructions for setting up the development environment for our mobile application project. The guide assumes no prior experience with mobile development. By following these steps, you'll be able to quickly and efficiently set up the environment and start contributing to the project.

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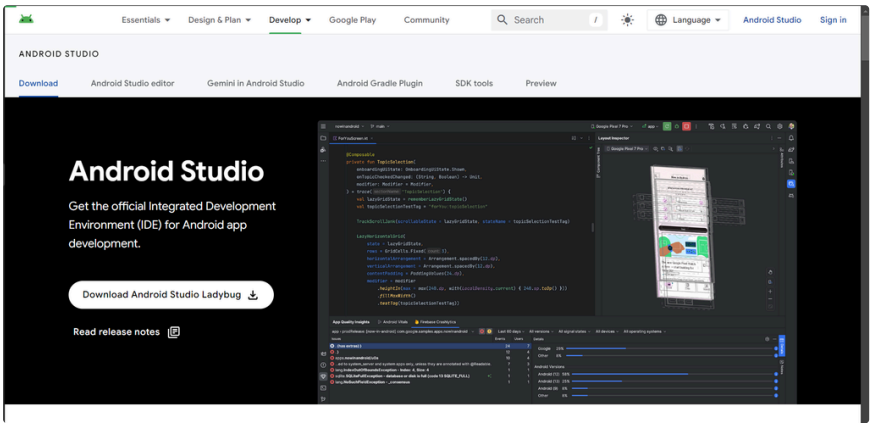
1. Required Software and Downloads

To work on this project, you'll need to install the following software:

1. **Android Studio (Ladybug or newer)**
   - *Download link* for tested version (Ladybug) **(RECOMMENDED)**
   - *Download page* for latest version
   - Follow the installation instructions on the download page.



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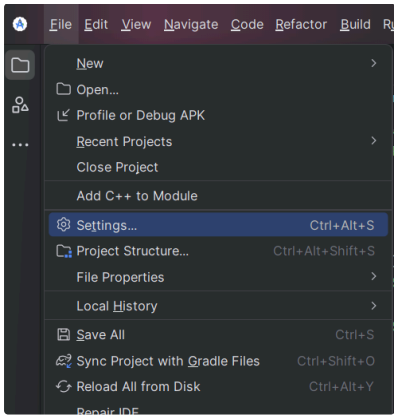
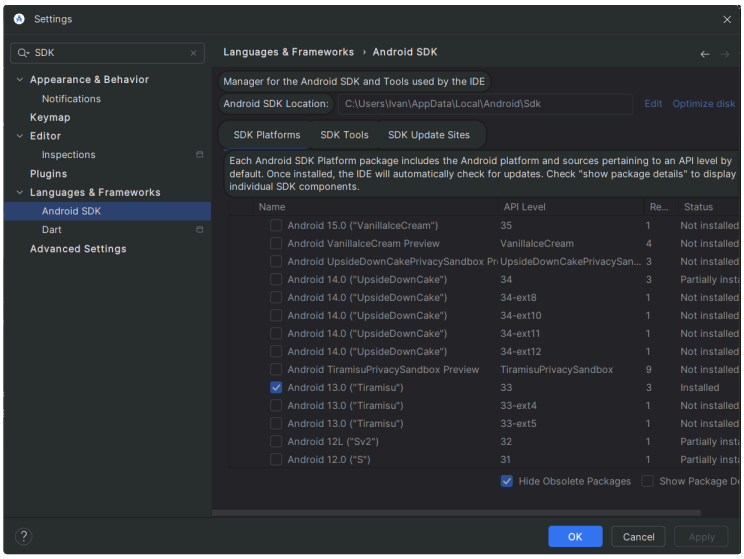
2. **Java Development Kit (JDK)**
   - Android Studio typically comes with an embedded JDK. Verify installation by going to *File > Project Structure > SDK Location* and checking the JDK path. If necessary, you can manually download JDK 11.
   - Download page for JDK 11.
3. **Android SDK and Emulator**
   - After installing Android Studio, open it and go to *SDK Manager* (*File > Settings > Languages & Frameworks > Android SDK*) to install Android SDK and other essential components, including Android Emulator (for testing the app on a virtual device).
4. **Android SDK and Physical device**
   - It is **RECOMMENDED** to use a physical device, running an appropriate version of Android, instead of an emulator. It's much faster and more reliable.
   - There's two options of connecting your device; via cable or via WiFi. The former is **RECOMMENDED** as it is more reliable.

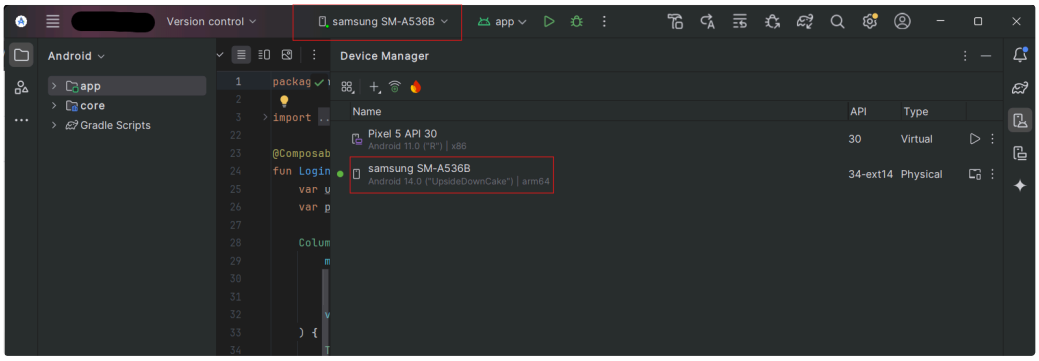
- Link for detailed instructions for physical device setup



3. Android SDK and Emulator



3. Android SDK and Emulator



4. Android SDK and Physical device

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2. Project Setup

Once Android Studio is installed, follow these steps to set up the project:

1. **Clone the Project Repository**
   - Open Android Studio, select *File > New > Project from Version Control*, and paste the repository link.

2. **Sync Project with Gradle Files**
   - After cloning, open the project and let Gradle sync. This may take a few minutes as dependencies will automatically be downloaded.

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3. Dependencies

The project uses several libraries to support development. Most dependencies will be **automatically** downloaded by Gradle, but here's a summary of what's included:

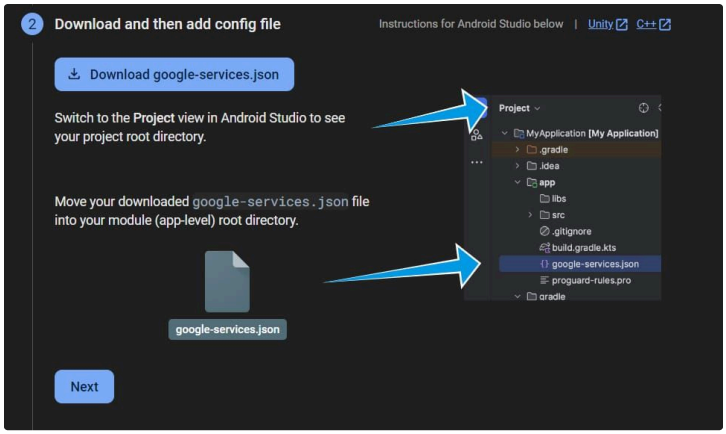
- **Kotlin Language**: Primary programming language used for the app.
- **Jetpack Compose**: UI toolkit for building native Android interfaces.
- **Material 3**: Design system used for a modern, intuitive interface.
- **GSON**: Library for converting JSON data to Kotlin objects.
- **Retrofit and OkHttpClient**: For managing network requests.

Below is a list of the primary dependencies specified in the project's `build.gradle` file:

```
// Material Design
implementation 'androidx.compose.material3:material3:<latest_version>'

// Navigation
def nav_version = "2.7.4"
implementation "androidx.navigation:navigation-compose:$nav_version"

// JSON Parsing
implementation 'com.google.code.gson:gson:2.9.0'

// Network Requests
implementation 'com.squareup.retrofit2:retrofit:2.2.0'
implementation 'com.squareup.okhttp3:okhttp:4.12.0'
implementation 'com.squareup.retrofit2:converter-gson:2.2.0'

// Core Android KTX
implementation 'androidx.core:core-ktx:1.9.0'

//FIREBASE
implementation(platform("com.google.firebase:firebase-bom:33.5.1"))
implementation("com.google.firebase:firebase-analytics")
implementation("com.google.firebase:firebase-auth")
implementation("com.google.firebase:firebase-firestore")
```

Note 1: `androidx.core:core-ktx` is included here to extend Android APIs with Kotlin extensions and is recommended for ease of use.

Note 2: `google-services.json` is not included in the original git. You need to manually add it into the project. To to this, put the file into the 'app' folder of the project (directly in the folder, not in a subfolder). The download for the file is below. ← **VERY IMPORTANT**



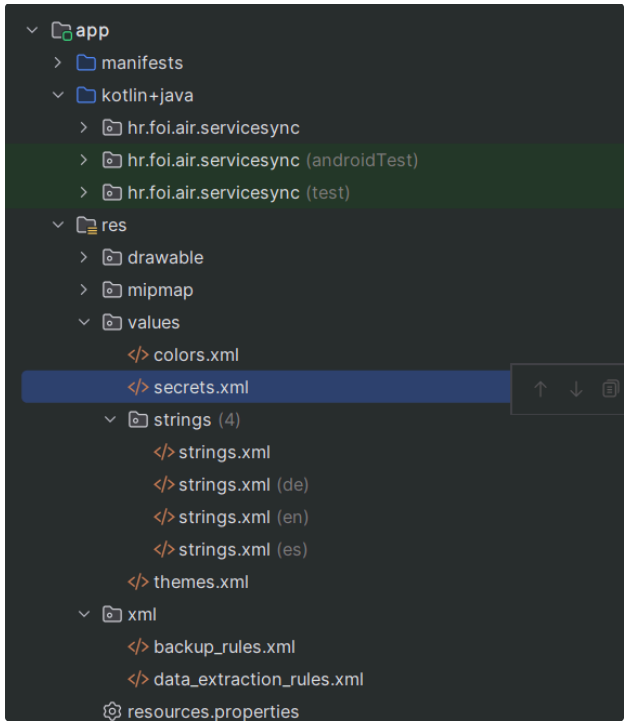
Note 2

google-services.json
10 Nov 2024, 03:14 PM

Note 3: `local.properties` does not have Google Maps API key. It has to be added manually. → MAPS_API_KEY=AIzaSyBPNdAVwD1wKD_6l9jZQ3N7VGcW0PAd9q4

Copy this into a new line (if you have other variables in local.properties, they can remain) ← **ALSO VERY IMPORTANT**

Note 4: Imgur API key is not in the project. File (provided below) has to be added manually into `res/values` folder that is located in the "app" module as shown in picture:



Location of the secrets.xml file

secrets.xml
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Please add this file or else profile image change feature won't work. ← **ALSO VERY IMPORTANT**

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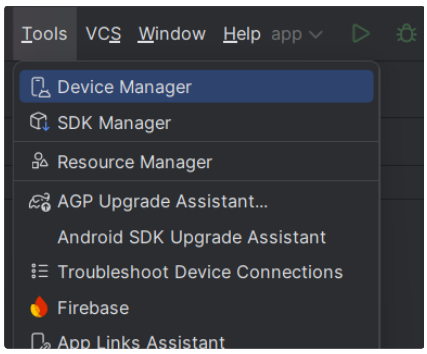
4. Setting up android emulator (or use a physical device)

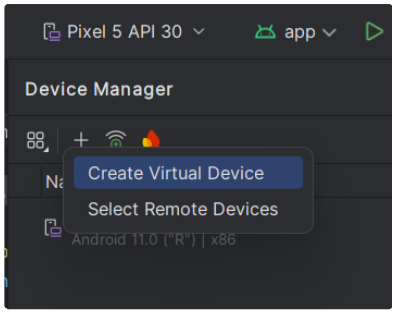
To test the application, you'll need an emulator:

1. **Open Device Manager**: In Android Studio, go to *Tools > Device Manager* and select *Create Virtual Device*.
2. **Select a device**: Choose a device profile, such as *Pixel 5*.
3. **Install system image**: Select the latest Android system image and finish the setup.
4. **Run the emulator**: Start the emulator from Device Manager.

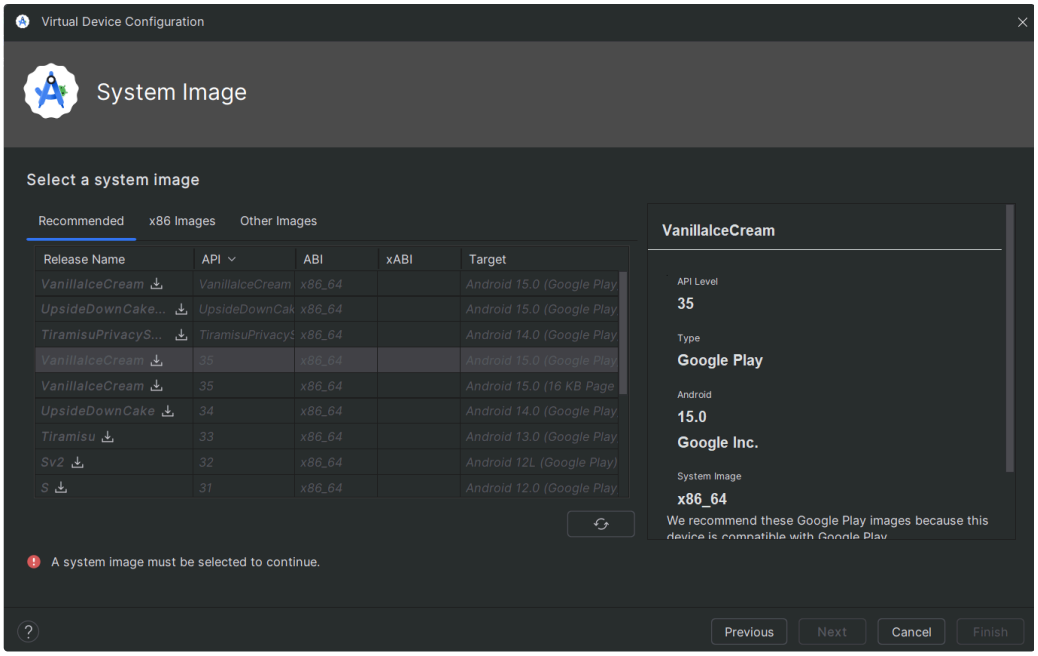
As it is already said, it's **RECOMMENDED** to use a physical device for faster and more reliable use of the device.



1. Open device manager



2. Select a device



3. Install system image

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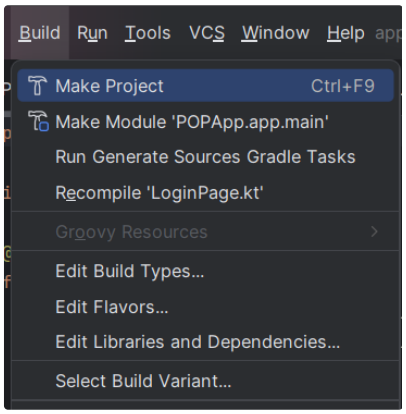
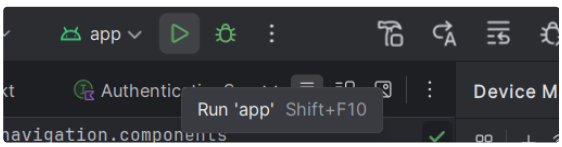
5. Building and Running the App

1. **Build the project**: Use the *Build > Make Project* option in Android Studio to build the app.
2. **Run the app**: Select the *Run* button or press *Shift + F10* to start the app on your selected emulator or connected device.



1. Build the project



2. Run the app

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6. Additional IMPORTANT things

System diagrams such as Class and ERA are here.

Team documentation is here.

Email + password for testing the Android app are: dpetek21@foi.hr, 123456

Registration is also available if you want to make your own account, but mind that user details on the profile page are blank for newly registered users and need to be edited and saved via ServiceSync app.

Details on how we implemented modular maps are here

How to contribute to the project by adding new map providers to the app is here

Details on how to run the notification server are here

Details on how the notification server is implemented are here

Adding additional companies by a registered user is possible using a web form provided here → `addCompany.html`

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7. Additional Resources

- Jetpack Compose Documentation
- Material Design Guidelines
- GSON Library Documentation
- Retrofit Documentation
- OkHttp Documentation

Each of these resources provides in-depth guidance for specific libraries and tools used in this project.

This guide should enable a smooth setup for developers, even if they are new to mobile development. If there are any issues, please reach out to the project administrator for support.